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Emergency departments serve as a critical entry point into the healthcare system, handling everything from minor injuries to life-threatening conditions. Because patient demand varies widely across regions and conditions, hospitals must continuously balance resources against need. This analysis explores how emergency department burden—measured as the ratio of patient visits to available treatment stations—varies across health conditions and over time. The data will be coming from the Emergency Department (ED) Volume and Capacity dataset, available through the U.S. government open data portal, and covers 2021 through 2023 across 12,849 observations and 18 variables. After cleaning to suit this analysis, the new dataset focuses on six key variables: year, county, health condition category, total ED visits, number of treatment stations, and visits per station. The central variable of interest, `Visits_Per_Station`, captures how hard each station is working—a higher value means more patients per station, which signals greater strain on resources.

The first visualization examines the overall distribution of `Visits_Per_Station`. The histogram reveals a right-skewed distribution: most emergency departments fall within a moderate range, but a meaningful tail extends toward much higher values. This pattern suggests that while the majority of facilities maintain a manageable ratio of patients to stations, a smaller subset operates under considerably more pressure. These high-burden outliers likely reflect a combination of factors: dense urban populations, limited resources, or both. It is worth noting

that the skew itself is informative, telling us that ED burden is not uniform, and policies aimed at reducing strain may need to be targeted rather than broad.

The second visualization compares burden across health condition categories using a boxplot. Some conditions are clearly associated with higher ED demand relative to capacity. Hypertension, mental health conditions, diabetes, and substance abuse all show elevated median values and wide distributions—suggesting these conditions place consistent, outsized pressure on emergency departments. By contrast, conditions like stroke, sepsis, and cardiac events show lower median burden. This may seem counterintuitive given their medical severity, but these conditions may simply arise less frequently, resulting in lower aggregate strain on treatment stations. The presence of many outliers across categories points to substantial hospital-to-hospital variability, which likely reflects differences in regional demographics, local health trends, and access to alternative care options.

The third visualization tracks average Visits\_Per\_Station across 2021, 2022, and 2023. The trend line is notably flat: there is a slight dip from 2021 to 2022, followed by a modest uptick in 2023, but the year-to-year differences are small. This stability suggests that, at the system level, hospitals largely kept pace with fluctuating patient demand—possibly through operational adjustments or resource reallocation. That said, three years of data is a limited window, and it remains unclear whether this stability reflects a genuine long-term equilibrium or simply a snapshot of a longer, more complex trajectory.

The final visualization uses a scatterplot to examine the relationship between treatment station count and total ED visits. As expected, there is a positive association: hospitals with more stations tend to see more patients. Larger facilities appear to be built for higher volume. However, the relationship is far from perfect. Hospitals with similar station counts often show

very different visit totals, which suggests that capacity alone does not determine demand.

Population density, proximity to other healthcare facilities, and hospital specialization all likely play a role in shaping how many patients a given ED serves.

Taken together, these findings offer a clearer picture of where ED burden is concentrated. Conditions like hypertension, mental health, diabetes, and substance abuse appear to contribute disproportionately to emergency department strain, while overall system-level burden remained relatively stable between 2021 and 2023. Several limitations are worth acknowledging: the dataset aggregates visits by condition category rather than individual patient records, condition classification may vary across hospitals, and the three-year window limits conclusions about longer-term trends. Still, this analysis provides a useful starting point for thinking about how healthcare systems can better anticipate and manage demand—particularly for the chronic and behavioral health conditions that seem to drive the heaviest ED workload.